

1. (10 points) **If a face image is a 100×100 image, written in row-major, this is a 10,000-dimensional vector. If we shift the image one pixel to the right, this will be a very different vector in the 10,000-dimensional space. How can we build face recognizers robust to such distortions?**

Instead of treating the set of pixels as dimensions in our data, we can instead extract information directly from the image such as distant between eyes, size of iris, skin tone, hair length, whether beard is present, etc. These features do not rely on spatial location of the pixels.

2. (10 points) **Suppose we have some binary data, $x_i \in \{0, 1\}$. The training data is as follows:**

x	t
0	$[-1, 1]^T$
0	$[-1, -2]^T$
0	$[-1, -1]^T$
1	$[1, 1]^T$
1	$[1, 2]^T$
1	$[2, 1]^T$

Let us embed each x_i into a 2D feature space using the basis function:

$$\phi(0) = [1, 0]^T, \quad \phi(1) = [0, 1]^T$$

Consider the multiple¹ linear regression model $y = \mathbf{W}\phi(x)$, where $\mathbf{W}$ is a 2×2 matrix. Compute the Maximum Likelihood Estimation (MLE) for $\mathbf{W}$ from the above data. The MLE solution for $\mathbf{W}$ is

$$\mathbf{W} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{t}$$

Based on the data provided in the table, the feature matrix $\mathbf{X}$ and the target matrix are defined as

$$\mathbf{X} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{t} = \begin{bmatrix} -1 & 1 \\ -1 & -2 \\ -1 & -1 \\ 1 & 1 \\ 1 & 2 \\ 2 & 1 \end{bmatrix}$$

¹Multiple linear regression refers to situations in which the model predicts more than one variable.

Putting it all together we find:

$$\begin{aligned}
 \mathbf{W} &= \left(\begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}^T \begin{bmatrix} -1 & 1 \\ -1 & -2 \\ -1 & -1 \\ 1 & 1 \\ 1 & 2 \\ 2 & 1 \end{bmatrix} \\
 &= \left(\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ -1 & -2 \\ -1 & -1 \\ 1 & 1 \\ 1 & 2 \\ 2 & 1 \end{bmatrix} \\
 &= \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ -1 & -2 \\ -1 & -1 \\ 1 & 1 \\ 1 & 2 \\ 2 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} -1 & -2/3 \\ 4/3 & 4/3 \end{bmatrix}
 \end{aligned}$$

3. (15 points) **Consider a training dataset $\{(x_i, t_i)\}_{i=1}^N$ where $x_i \in \mathbb{R}^D$, $t_i \in \mathbb{R}$ and D is the feature space dimensionality. Suppose you are training a linear regression model, $y(x) = \mathbf{w}^T \mathbf{x} + w_0$. Answer the following questions:**

(a) (4 points) **Consider the objective function**

$$J(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N (t_i - y(x_i))^2 + \lambda R(\mathbf{w})$$

What is the role of the regularizer $R(\mathbf{w})$? What is the effect of changing the value of λ ?

The regularizer was introduced to mitigate overfitting when training the model because it constraints the values of the parameter $\mathbf{w}$ to be small. The most common regularizers (covered in class) are:

- the Ridge regularizer $R_{L_2}(\mathbf{w}) = \|\mathbf{w}\|_2^2$. It drives all coefficients in the vector $\mathbf{w}$ to be small-valued and it drastically penalizes large values.
- the Lasso regularizer $R_{L_1}(\mathbf{w}) = \|\mathbf{w}\|_1$. It drives some of the coefficients to exactly zero and thus it's a great tool for mitigating overfitting, the curse of dimensionality and feature selection. It does not penalize outliers as drastically as Ridge.

- the Elastic Net regularizer $R_{L_{1,2}}(\mathbf{w}) = \beta R_{L_1}(\mathbf{w}) + (1 - \beta)R_{L_2}(\mathbf{w})$. It includes both Lasso and Ridge regularizers.

The role of λ is to control the trade-off between fitting the curve to the training data (minimizing the error) and avoiding overfitting. A large value for λ will shrink the coefficient values to zero and underfit the data. A small value of λ may lead to overfitting.

- (b) (4 points) **How would you train a linear regression model that is robust to outliers?**

Instead of using the Least Squares objective function, we could use the Mean Absolute Error (MAE). Using the absolute error mitigates the influence of large errors and becomes more robust to outliers.

- (c) (4 points) **In scenarios where at least two dimensions exhibit collinearity, meaning they are linearly dependent, which regularization technique would you employ in your linear regression model while preserving all dimensions?**

When 2 dimensions are dependent, then generally we would not want to use both of them. Lasso regularizer would be a good regularization term in this scenario as it would be able to identify features that do not contribute to the target predictions.

- (d) (3 points) **How does regularization help address the challenges posed by the curse of dimensionality?**

Some regularizers, like Lasso and Elastic Net, help select a subset of features useful for model prediction, and thus reducing the dimensionality of the feature space.

4. (10 points) **In the field of financial transactions, such as online retail, when predicting the total cost of an online purchase, errors in the prediction are likely to be strictly positive because the actual cost cannot be less than zero.**

Let $\epsilon_i = t_i - y_i$ represent the (positive) residual error for input sample x_i and consider the following objective function:

$$J(\mathbf{w}) = \sum_{i=1}^N \epsilon_i$$

What is the Bayesian interpretation of this objective function? Show and justify your work.

Since this objective function is a function of the (positive) residual error, we want to find the model parameters, θ , that minimize it. We have:

$$\begin{aligned}
\arg_{\theta} \min \sum_{i=1}^N \epsilon_i &= \arg_{\theta} \max - \sum_{i=1}^N \epsilon_i \\
&= \arg_{\theta} \max \exp \left\{ - \sum_{i=1}^N \epsilon_i \right\} \\
&= \arg_{\theta} \max \prod_{i=1}^N e^{-\epsilon_i} \\
&= \arg_{\theta} \max \prod_{i=1}^N \lambda e^{-\lambda \epsilon_i} \\
&\propto \arg_{\theta} \max \prod_{i=1}^N \text{Exp}(\epsilon_i | \lambda)
\end{aligned}$$

Thus, the residuals follow an Exponential distribution with parameter λ .

5. (10 points) **Suppose we are modeling coin tosses (Bernoulli trials),**

$$P(x|\mu) = \mu^x(1-\mu)^{1-x} \quad \text{where } x \in \{0, 1\}$$

and you think the coin (modeled by parameter μ) is either fair, or is biased towards heads. To model this kind of prior belief, you consider a mixture of two beta distributions:

$$P(\mu) = 0.5\text{Beta}(\mu|\alpha_1, \beta_1) + 0.5\text{Beta}(\mu|\alpha_2, \beta_2)$$

Answer the following questions:

- (a) (5 points) **Write down the observed data likelihood for the Maximum A Posteriori (MAP) approach.**

The observed data likelihood for the MAP approach is as follows:

$$\begin{aligned}
\mathcal{L}^0 &= P(X|\mu)P(\mu) \\
&= \left(\prod_{i=1}^N \mu^{x_i} (1-\mu)^{1-x_i} \right) (0.5\text{Beta}(\mu|\alpha_1, \beta_1) + 0.5\text{Beta}(\mu|\alpha_2, \beta_2)) \\
&= 0.5 \left(\prod_{i=1}^N \mu^{x_i} (1-\mu)^{1-x_i} \right) \text{Beta}(\mu|\alpha_1, \beta_1) + 0.5 \left(\prod_{i=1}^N \mu^{x_i} (1-\mu)^{1-x_i} \right) \text{Beta}(\mu|\alpha_2, \beta_2) \\
&\propto 0.5 \mu^{\sum_{i=1}^N x_i + \alpha_1 - 1} (1-\mu)^{N - \sum_{i=1}^N x_i + \beta_1 - 1} + 0.5 \mu^{\sum_{i=1}^N x_i + \alpha_2 - 1} (1-\mu)^{N - \sum_{i=1}^N x_i + \beta_2 - 1} \\
&\propto 0.5 \text{Beta} \left(\mu \left| \sum_{i=1}^N x_i + \alpha_1, N - \sum_{i=1}^N x_i + \beta_1 \right. \right) + 0.5 \text{Beta} \left(\mu \left| \sum_{i=1}^N x_i + \alpha_2, N - \sum_{i=1}^N x_i + \beta_2 \right. \right)
\end{aligned}$$

- (b) (5 points) **Is this mixture model a conjugate prior? Why or why not?**

From the observed likelihood above, we find it to also follow a mixture of two Beta distributions with the same weight responsibilities (0.5 each). Thus this mixture model forms a conjugate prior.

6. (15 points) **Consider a mixture model with 2 components ($K = 2$), where one component is Gaussian-distributed, $\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$, and the other component is Exponential-distributed, $E(x|\lambda) = \lambda e^{-\lambda x}$. For a dataset with N i.i.d. samples $\{x_i\}_{i=1}^N$, answer the following questions:**

- (a) (8 points) **Introduce the hidden latent variables for this mixture model and write down the complete data likelihood.**

The latent variables z_i corresponds to the label of the component from which sample x_i was drawn from. Since we only have 2 components, $z_i \in \{0, 1\}$. The complete data likelihood can be written as:

$$\begin{aligned}\mathcal{L}^c &= \prod_{i=1}^N [\pi_{z_i} P_{z_i}(x_i|\Theta)]^{z_i}, \quad z_i \in \{0, 1\} \\ &= \prod_{i=1}^N [\pi_1 \mathcal{N}(x_i|\mu, \sigma^2)]^{z_i} [\pi_0 E(x_i|\lambda)]^{1-z_i} \\ &= \prod_{i=1}^N \left[\pi_1 (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right) \right]^{z_i} [\pi_0 \lambda e^{-\lambda x_i}]^{1-z_i}\end{aligned}$$

where $0 \leq \pi_i \leq 1, i = 0, 1, \pi_0 + \pi_1 = 1$ and $\Theta = \{\pi_0, \pi_1, \mu, \sigma^2, \lambda\}$. Here, I'm assigning $z = 1$ for the Gaussian component, and $z = 0$ for the Exponential component.

- (b) (7 points) **Derive the update equation for the parameter λ .** The Q-function is defined as:

$$\begin{aligned}Q(\Theta, \Theta^{(t)}) &= \mathbb{E}[\ln \mathcal{L}^c] \\ &= \sum_{k=0}^1 \ln \mathcal{L}^c P(z_i = k|x_i, \Theta^{(t)}) \\ &= \sum_{k=0}^1 \ln \mathcal{L}^c C_{ik}\end{aligned}$$

Optimizing for the parameters in the mixture model corresponds to the M-STEP, where the memberships values are fixed. Thus, to solve for λ we take the derivative of $Q(\Theta, \Theta^{(t)})$ with respect to λ (MLE solution).

$$\begin{aligned}
Q(\Theta, \Theta^{(t)}) &= \mathbb{E} [\ln \mathcal{L}^c] \\
&= \sum_{i=1}^N \sum_{k=0}^1 \ln \mathcal{L}^c P(z_i = k | x_i, \Theta^{(t)}) \\
&= \sum_{i=1}^N \sum_{k=0}^1 \ln \mathcal{L}^c C_{ik}
\end{aligned}$$

Optimizing for the parameters in the mixture model corresponds to the M-STEP, where the memberships values are fixed. Thus, to solve for λ we take the derivative of $Q(\Theta, \Theta^{(t)})$ with respect to λ (MLE solution).

$$Q(\Theta, \Theta^{(t)}) = \sum_{i=1}^N \left[z_i \left(\ln \pi_1 - \frac{1}{2}(2\pi\sigma^2) - \frac{(x_i - \mu)^2}{2\sigma^2} \right) C_{i1} + (1 - z_i) (\ln \pi_2 + \ln \lambda - \lambda x_i) C_{i0} \right]$$

$$\begin{aligned}
\frac{\partial Q(\Theta, \Theta^{(t)})}{\partial \lambda} = 0 &\iff \sum_{i=1}^N (1 - z_i) \left(\frac{1}{\lambda} - x_i \right) C_{i0} = 0 \\
&\iff \sum_{i=1}^N \left(\frac{1}{\lambda} - x_i - \frac{z_i}{\lambda} + z_i x_i \right) C_{i0} = 0 \\
&\iff \sum_{i=1}^N (1 - \lambda x_i - z_i + \lambda z_i x_i) C_{i0} = 0 \\
&\iff \sum_{i=1}^N (1 - z_i) C_{i0} - \lambda \sum_{i=1}^N (x_i - z_i x_i) C_{i0} = 0 \\
&\iff \lambda = \frac{\sum_{i=1}^N (1 - z_i) C_{i0}}{\sum_{i=1}^N (x_i - z_i x_i) C_{i0}} \\
&\iff \lambda = \frac{\sum_{\{x_i: z_i=0\}} C_{i0}}{\sum_{\{x_i: z_i=0\}} x_i C_{i0}}
\end{aligned}$$

7. (10 points) **Consider a binary classification task, let the positive class be C_1 and negative class C_0 . A positive sample occurs with probability $p_1 = P(C_1)$, and a negative sample with probability $p_0 = P(C_0)$.**

Suppose that you have determined that each class is modeled according to a bivariate Gaussian distribution of the form:

$$P(\mathbf{x}|C_1) \sim \mathcal{N} \left(\mu_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \Sigma_1 = \mathbf{I} \right) \quad \text{and} \quad P(\mathbf{x}|C_0) \sim \mathcal{N} \left(\mu_2 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \Sigma_2 = \mathbf{I} \right)$$

where $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $\mathbf{I}$ is the 2×2 identity matrix.

- (a) (6 points) **Let $g_i(\mathbf{x}) = \ln(P(\mathbf{x}|C_i)P(C_i))$ be the discriminant function for class C_i . Then, let $g(\mathbf{x}) = g_1(\mathbf{x}) - g_0(\mathbf{x})$ be the Naïve Bayes decision boundary. Find $g(\mathbf{x})$ for this example. Show your work.**

To compute the data likelihoods, we need to determine the inverse of the covariance matrix. Since they are both the identity matrix $\Sigma_1 = \Sigma_2 = \mathbf{I}$, we know that $\Sigma_1^{-1} = \Sigma_2^{-1} = \mathbf{I}^{-1} = \mathbf{I}$.

The discriminant function $g_i(x)$, $i = 1, 2$ is defined as $P(x|C_i)P(C_i)$. We can simplify it by taking $g_i(x) = \ln(P(x|C_i)P(C_i))$. Note that $x = [x_1, x_2]^T$. We have:

$$\begin{aligned} g_1(x) &= \ln \left(p_1 \times \frac{1}{(2\pi)^{2/2}|\Sigma_1|^{1/2}} \exp \left\{ -\frac{1}{2} \begin{bmatrix} x_1 - 1 \\ x_2 \end{bmatrix}^T \Sigma_1^{-1} \begin{bmatrix} x_1 - 1 \\ x_2 \end{bmatrix} \right\} \right) \\ &= \ln(p_1) - \ln(2\pi) - \frac{1}{2}((x_1 - 1)^2 + x_2^2) \\ &= \ln(p_1) - \ln(2\pi) - \frac{1}{2}(x_1^2 - 2x_1 + 1 + x_2^2) \end{aligned}$$

and

$$\begin{aligned} g_0(x) &= \ln \left(p_0 \times \frac{1}{(2\pi)^{2/2}|\Sigma_2|^{1/2}} \exp \left\{ -\frac{1}{2} \begin{bmatrix} x_1 \\ x_2 + 1 \end{bmatrix}^T \Sigma_2^{-1} \begin{bmatrix} x_1 \\ x_2 + 1 \end{bmatrix} \right\} \right) \\ &= \ln(p_0) - \ln(2\pi) - \frac{1}{2}(x_1^2 + (x_2 + 1)^2) \\ &= \ln(p_0) - \ln(2\pi) - \frac{1}{2}(x_1^2 + x_2^2 + 2x_2 + 1) \end{aligned}$$

Now, let $g(x) = g_1(x) - g_0(x)$, we find:

$$\begin{aligned} g(x) &= -\frac{1}{2}(x_1^2 - 2x_1 + 1 + x_2^2) + \frac{1}{2}(x_1^2 + x_2^2 + 2x_2 + 1) + \ln \left(\frac{p_1}{p_0} \right) \\ &= x_1 + x_2 + \ln \left(\frac{p_1}{p_0} \right) \end{aligned}$$

Thus, we decide C_1 (label 1) if $g(x) > 0 \iff x_1 + x_2 + \ln \left(\frac{p_1}{p_0} \right) > 0$.

- (b) (4 points) **Let $P(C_1) = \frac{1}{5}$. In the table below, include the label prediction (0 or 1) for each sample (x_1, x_2) based on your solution for part (a). Show your work. (Useful information: $\ln(1/4) \approx -1.3862$.)**

Since we have a 2-class problem and $p_1 = \frac{1}{5}$, then $p_0 = 1 - p_1 = \frac{4}{5}$. Thus $\ln \left(\frac{p_1}{p_0} \right) = \ln \left(\frac{1}{4} \right) \approx -1.3862$.

Input $x_i = (x_i^{(1)}, x_i^{(2)})$	Classifier Prediction y_i
(0, 0)	$0 + 0 - 1.3862 < 0 \Rightarrow \text{predict } 0$
(1, 0)	$1 + 0 - 1.3862 < 0 \Rightarrow \text{predict } 0$
(2, 2)	$2 + 2 - 1.3862 > 0 \Rightarrow \text{predict } 1$
(-1, 2)	$-1 + 2 - 1.3862 < 0 \Rightarrow \text{predict } 0$

8. (10 points) **Within clustering algorithms, we discussed Gaussian Mixture Models (GMMs) and the K-Means algorithm. What are the advantages of GMMs over K-Means? Describe at least 4 unique advantages.**

GMMs have the following advantages over k-means clustering:

1. Cluster density. GMMs are able to converge to clusters with varying cluster sizes, whereas k-means tends to merge a small group of samples with the nearest dense cluster.
2. Outliers. GMMs is less sensitive to outliers because it uses probability densities as its similarity measure instead of distance-based similarity.
3. Soft Membership. GMMs use soft membership assignment. This uncertainty level allows us to further select which samples to consider in any particular cluster based on a probability threshold.
4. Cluster shapes. GMMs allows each cluster to have a varying shape (spherical or ellipsoidal), whereas, in k-means, all clusters will have the same shape (e.g. spherical with Euclidean distance, ellipsoidal with Mahalanobis distance).

Despite of the obvious advantages of GMMs over k-means, GMMs are more computationally expensive.

9. (10 points) **One source of noise is error in the labels. Using tools introduced in lecture, can you propose a method to find data points that are highly likely to be misclassified?**

If we have an instance that is surrounded by many instances with a different label, then it is most probably mislabeled. By using a neighbor-based method like K-Nearest Neighbors (KNN), we can check for this. The steps would go as follows:

- Train a KNN model on the training dataset using the available labels.
- Use the trained KNN model to predict the labels for each sample in the dataset.
- For each sample prediction, calculate the confidence score based on the proportion of neighbors with the same predicted label.
- Define a confidence threshold below which predictions are considered uncertain.
- Identify sample where the confidence score is below the threshold. These samples are likely to be misclassified or have uncertain labels.